

5(a)(i)	loss of (signal) power/amplitude/intensity	B1
5(a)(ii)	unwanted/random signal	B1
	superposed on (transmitted) signal	B1
5(b)(i)	attenuation = $10 \lg(P_2 / P_1)$	C1
	attenuation per unit length = $(1 / L) \times 10 \lg(P_2 / P_1)$ $= (1 / 52) \times 10 \lg [(2.5 \times 10^{-3}) / (7.8 \times 10^{-16})]$	C1
	$= 2.4 \text{ dB km}^{-1}$	A1
5(b)(ii)	gain / dB = $10 \lg(P_2 / P_1)$ $115 = 10 \lg [P / (7.8 \times 10^{-16})]$	C1
	$P = 2.5 \times 10^{-4} \text{ W}$	A1